

SparseNav: Instruction-conditioned Sparse Semantic Perception for Training-Free Vision-Language Navigation

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Abstract—Map-based vision-language navigation (VLN) relies on persistent spatial representations to connect language understanding with geometric planning. However, acquiring semantics beyond the needs of the current instruction can introduce unnecessary perception cost and irrelevant annotations. Continuously accumulating unrelated objects may not only waste computation, but also clutter the visual-spatial representation consumed by the vision-language model (VLM) planner. To address this problem, we present *SparseNav*, a training-free framework that follows a less-is-more principle for semantic navigation. *SparseNav* persistently maintains a lightweight geometric bird’s-eye-view (BEV) map and sparse landmark memory, acquiring new semantics on demand using the active sub-instruction to decide what is worth grounding. An instruction manager first tracks navigation progress and identifies the active landmark query. An instruction-conditioned perception mechanism then invokes open-vocabulary segmentation when the queried landmark is visible and its metric location can inform the next decision. The resulting landmark memory supports VLM selection among hybrid frontier and local directional waypoint candidates. Without any additional training, *SparseNav* achieves success rates of 42.8% on R2R-CE and 40.7% on RxR-CE, both on the Val-Unseen splits. Controlled ablations examine semantic perception strategies and the contributions of individual framework components. Furthermore, we successfully deployed *SparseNav* on a Unitree Go2 quadruped equipped with an Intel RealSense D455 RGB-D camera for geometric mapping and landmark grounding and a Livox MID-360 LiDAR for localization, without a prebuilt map. We validated its effectiveness across multiple indoor environments using instruction-conditioned waypoint navigation.

I. INTRODUCTION

Vision-language navigation (VLN) (e.g., [1], [2]) aims to enable an embodied agent to navigate through an environment according to natural-language instructions. Given a language instruction, VLN requires the agent to jointly perform visual perception, language grounding, spatial reasoning, exploration, progress tracking, and low-level motion execution. Executing such instructions therefore requires a consistent connection between semantic understanding and geometric planning.

The recent emergence of large vision-language models (VLMs) (e.g., [3], [4], [5], [6], [7], [8], [9]) has substantially improved general visual and linguistic reasoning. Their open-world knowledge makes them particularly attractive for

training-free VLN, where navigation-specific demonstrations or task-specific model training can be avoided. However, directly asking a VLM to predict low-level robot actions from egocentric images remains difficult. This difficulty reflects three requirements that general-purpose visual reasoning alone does not fulfill. First, image-space reasoning does not by itself provide a persistent metric representation of traversable space. Furthermore, long navigation trajectories require persistent spatial memory that cannot be reliably represented by an ever-growing sequence of historical RGB images. Finally, natural-language instructions frequently refer to concrete landmarks whose approximate visual recognition is insufficient for precise navigation.

To meet these requirements, map-based navigation (e.g., [10], [11], [12], [13], [14]) provides a natural interface between high-level semantic reasoning and physical robot motion. A bird’s-eye-view representation explicitly exposes free space, obstacles, explored regions, robot pose, and candidate destinations. Recent methods have therefore introduced occupancy maps, value maps, annotated semantic maps, and hierarchical semantic-geometric maps into VLM-based navigation. These representations establish the value of persistent spatial memory, but leave open how much semantic information should be acquired for each navigation decision. We investigate whether maintaining broader semantic coverage necessarily improves instruction following.

Instruction-conditioned planning and sub-instruction tracking [15], [16] motivate our hypothesis that only a small subset of scene semantics needs to be explicitly grounded for the active navigation objective. Continuously accumulating unrelated objects may not only waste computation, but also clutter the visual-spatial representation consumed by the VLM planner. For example, consider the instruction fragment in Fig. 1. Language already specifies the relevant landmark pair: the *couches* and *kitchen counter*. Other visible objects need not be semantically annotated to resolve this relation. Their geometry can still be retained for obstacle avoidance. Segmenting every chair, lamp, cabinet, painting, and table requires additional computation and may introduce noisy detections and visual-map clutter without necessarily improving the navigation decision. As illustrated in Fig. 1, *SparseNav* therefore follows a **less-is-more** principle for semantic navigation: *Persist geometry, but perceive, ground, and remember semantics only when language makes them*

relevant.

In this work, we instantiate this principle through an agent-centered, heading-up geometric BEV, hybrid frontier/local waypoint candidates, and sparse landmark memory. The active sub-instruction determines the semantic query. A VLM invokes segmentation only when the queried landmark is visible and metric grounding can affect waypoint selection. Mask-depth projection supplies landmark centers for the BEV, and an instruction manager tracks completed and pending objectives.

Our contributions are summarized as follows:

- **Instruction-conditioned sparse semantic perception.** We introduce a semantic acquisition policy in which the active sub-instruction specifies the landmark queries, while a VLM-based trigger determines when metric grounding is needed.
- **A unified spatial interface for training-free VLN.** We integrate persistent geometry, sparse landmark memory, recent observations, and instruction progress into a BEV-based decision interface. Geometrically generated frontier and local candidates support VLM waypoint selection and classical path planning.
- **Benchmark evaluation and real-robot deployment.** We evaluate *SparseNav* on R2R-CE and RxR-CE and conduct controlled ablations of semantic perception strategies and key framework components. We deployed *SparseNav* on a legged robot and validated its effectiveness across multiple environments.

II. RELATED WORK

A. Vision-Language Navigation in Continuous Environments

VLN was originally formulated using discrete navigation graphs [1], while VLN-CE [2] extends the task to continuous environments where an agent must execute low-level actions rather than select transitions between predefined viewpoints. Continuous VLN introduces additional challenges including collision avoidance, metric localization, waypoint selection, and control.

Traditional approaches (e.g., [1], [2], [17], [18], [19]) learn navigation policies from human demonstrations or simulator-generated trajectories. While supervised methods have achieved strong benchmark performance, collecting navigation data is expensive and learned policies may have difficulty generalizing to instructions or visual environments that differ substantially from the training distribution. These requirements motivate complementary approaches that reuse pretrained models without navigation-specific training.

Recent work therefore explores training-free or zero-shot navigation using foundation models. InstructNav [15] introduces Dynamic Chain-of-Navigation reasoning and multi-source value maps for generic instruction navigation. CANav [16] reformulates navigation as sequential constraint-aware sub-instruction completion and explicitly maintains progress constraints. These methods demonstrate the feasibility of using pretrained models for navigation without navigation-specific training.

These approaches strongly motivate explicit spatial representations, but the value of persistent geometry does not establish that broader semantic coverage is always beneficial. *SparseNav* follows the training-free paradigm while explicitly controlling which landmarks are grounded and when their observations enter persistent semantic memory. This separates the value of persistent spatial memory from semantic density: geometry remains persistent, whereas semantic information is added only when requested by the active navigation instruction. The resulting map is deliberately incomplete with respect to the scene, and controlled ablations evaluate whether this selective representation preserves navigation quality.

B. Map-Based Memory for VLN

A persistent spatial representation is particularly valuable for long-horizon navigation because raw observation histories continuously increase in length and provide only implicit spatial relationships.

VLFM constructs occupancy maps from depth observations, identifies exploration frontiers, and combines these frontiers with vision-language semantic values for zero-shot ObjectNav [10]. This work demonstrates the effectiveness of combining geometric exploration with foundation-model semantics.

MapNav introduces an Annotated Semantic Map as a structured memory representation for VLM-based VLN, replacing long histories of raw visual observations with a top-down map containing explicit semantic annotations [11].

DreamNav [20] combines egocentric view correction, trajectory-level planning, and imagination-based prediction to support anticipatory, long-horizon zero-shot navigation.

SparseNav retains persistent geometry for traversability reasoning, while the active sub-instruction determines the landmark queries and the VLM selectively triggers their metric grounding. The previously grounded landmarks remain available in sparse memory, which makes semantic acquisition an explicit decision, allowing us to study whether instruction-relevant semantics can preserve navigation quality with fewer perception calls.

C. Open-Vocabulary Visual Grounding

Open-vocabulary detection and segmentation enable the grounding of natural-language concepts without predefined labels, accommodating varied landmark descriptions in VLN, such as “the couch next to the window.” Modern promptable segmentation models such as Segment Anything support prompt-conditioned mask prediction [21].

SAM 3 [22] extends this capability to concept prompts, including short noun phrases, thereby enabling text-conditioned detection and segmentation. In our implementation, *SparseNav* uses an open-vocabulary segmentation model as a callable visual perception tool.

SparseNav makes segmentation an **event-driven reasoning tool**: the current navigation state triggers perception only when the VLM identifies a potentially present and task-relevant landmark.

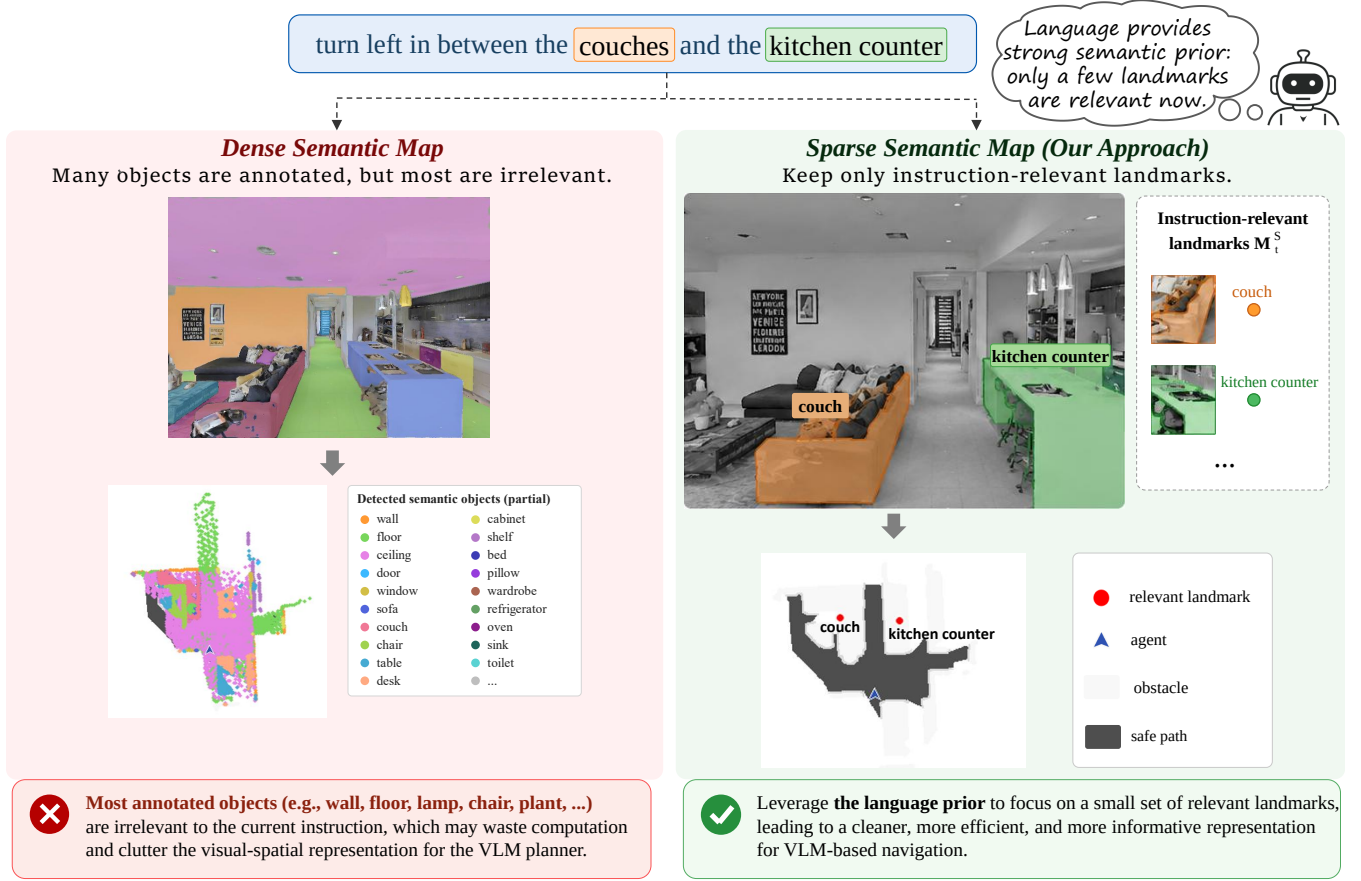


Fig. 1. Language provides a semantic prior for sparse navigation. For the instruction “turn left in between the couches and the kitchen counter,” dense annotation includes many objects unrelated to the current decision (left), whereas *SparseNav* grounds the relevant pair while retaining geometry (right). Sparse memory M_t^S stores landmark labels and cluster centers.

III. METHOD

A. Problem Formulation

Given instruction I and observations O_t , the agent selects a geometrically valid waypoint $p_t^* \in \mathcal{C}_t$ and executes it with a classical planner. The agent state is

$$\mathcal{M}_t = (M_t^G, M_t^S, E_t, q_t), \quad (1)$$

where M_t^G is persistent geometry, M_t^S is sparse landmark memory, E_t is recent visual history, and q_t is instruction progress. Fig. 2 summarizes the closed loop: the instruction manager supplies the active landmark query, RGB-D observations and odometry update M_t^G , and a VLM visibility check gates open-vocabulary segmentation. Mask-depth projection updates labels and cluster centers in M_t^S ; the VLM then selects from hybrid candidates using M_t^G , M_t^S , E_t , and q_t , while planning and execution feedback update subsequent decisions.

B. Geometry-Centric Spatial Memory

RGB-D or range observations are transformed to a common world frame. Points with $z_{\min} < z < z_{\max}$ are rasterized at resolution ρ to form an obstacle map. Elliptical dilation bridges nearby wall observations, and morphological closing merges fragmented regions and fills small holes:

$$B'_t = B_t \oplus K_{\text{ellipse}}, \quad B''_t = (B'_t \oplus K_c) \ominus K_c. \quad (2)$$

where B_t denotes the binary obstacle map before morphological post-processing.

Obstacles are inflated by $\lceil r_a/\rho \rceil$ pixels for the robot footprint, using $r_a = 0.1$ m; free-space accuracy is limited by observed geometry.

For VLM input, the map is translated and rotated to center the robot and align its heading upward:

$$p^{\text{ego}} = R(-\theta_t)(p - p_t). \quad (3)$$

The local frame uses $+x$ forward and $+y$ left. During rasterization, these axes map to image up and image left, respectively, with $u = u_0 - y_{\text{ego}}/\rho$ and $v = v_0 - x_{\text{ego}}/\rho$. Thus image directions match the linguistic notions of forward, left, and right.

C. Hybrid Exploration-Exploitation Waypoint Proposal

Frontiers discover unknown regions but miss useful waypoints in explored space, such as continuing forward before turning at a sofa.

SparseNav therefore defines a hybrid candidate set:

$$\mathcal{C}_t = \mathcal{C}_t^F \cup \mathcal{C}_t^L, \quad (4)$$

where $\mathcal{C}_t^F$ contains frontier candidates and $\mathcal{C}_t^L$ local directional candidates. Grounded landmarks guide selection among these two classes.

1) *Frontier Candidates*: Frontiers are navigable cells adjacent to unknown space; connected cells are clustered, small clusters rejected, and a reachable point with obstacle clearance represents each remaining cluster in $\mathcal{C}_t^F$.

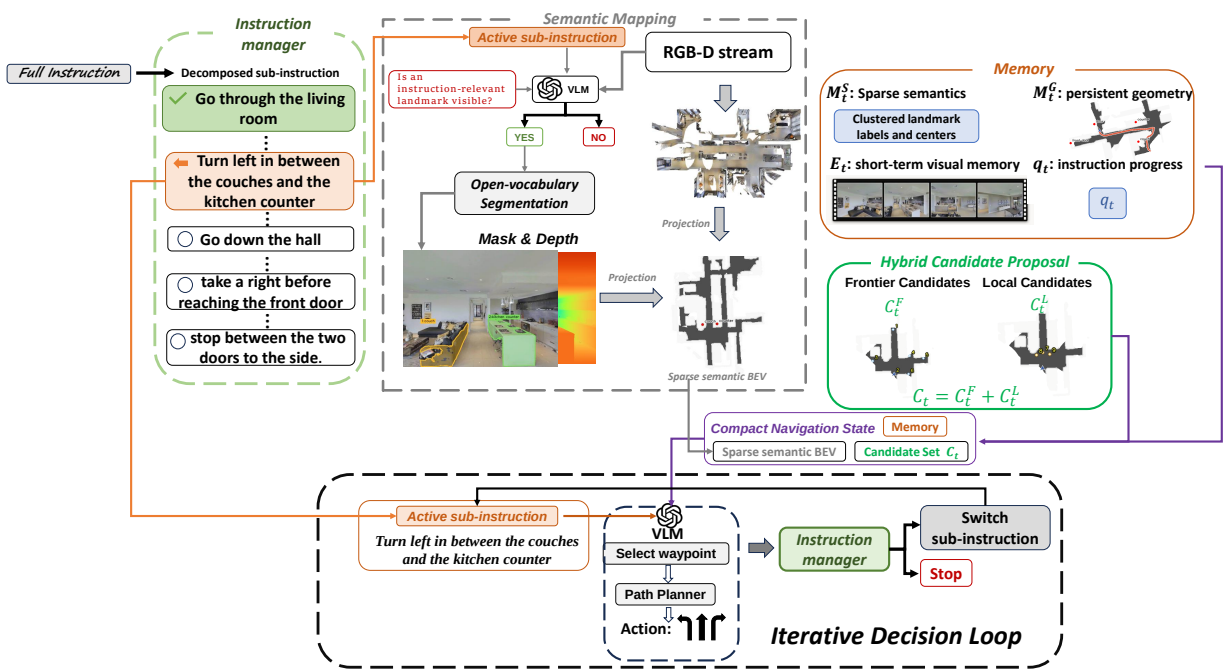


Fig. 2. The overview of *SparseNav*. The instruction manager maintains sub-instruction progress and an active landmark query. RGB-D observations and odometry update persistent geometry M_t^G ; a VLM visibility check gates open-vocabulary segmentation, and mask–depth projection updates landmark labels and cluster centers in M_t^S . These memories, recent observations E_t , progress q_t , and frontier/local candidates form the compact navigation state. The VLM selects a candidate, a path planner generates the trajectory, and execution feedback supports sub-instruction switching or stopping.

2) *Local Directional Candidates*: Local proposals preserve motion within explored space. For distance d_l and offsets $\phi_i \in \{-\phi, 0, +\phi\}$,

$$l_t^i = p_t + d_l[\cos(\theta_t + \phi_i), \sin(\theta_t + \phi_i)]^\top. \quad (5)$$

Candidates in inflated obstacles or disconnected regions are discarded or projected to free space; frontiers support discovery, while local left/front/right proposals support instruction-following maneuvers.

D. Hierarchical Instruction Management

The instruction manager decomposes I into ordered sub-instructions $S = (s_1, \dots, s_K)$ and maintains $q_t = (i_t, S_t^{\text{done}})$, where i_t indexes the active item and S_t^{done} records completed items. The VLM receives completed, current, and upcoming objectives. When the active completion condition is met, the manager records it and advances; progress also determines the active landmark query, avoiding repeated objectives while retaining upcoming context.

E. Instruction-Conditioned Sparse Semantic Perception

1) *Task-Relevant Landmark Query*: For active sub-instruction s_i , the instruction manager identifies an optional landmark query

$$l_i = \text{ExtractLandmark}(s_{i_t}, s_{i_t+1}, q_t). \quad (6)$$

This query can contain multiple labels, such as $l_i = \{\text{couch}, \text{kitchen counter}\}$ for a spatial relation. It is derived from the active sub-instruction; at a transition, the immediately upcoming landmark may also be activated when it constrains the next waypoint, without advancing q_t . Perception is restricted to this query rather than all visible objects.

2) *Semantic Trigger*: The VLM evaluates (O_t, E_t, s_i, l_i) and returns query labels, visibility scores, and a grounding decision $z_t \in \{0, 1\}$. Grounding is enabled when the landmark is likely visible and its metric position can affect the next decision; a temporal cooldown suppresses redundant calls unless prior localization is unreliable.

3) *Open-Vocabulary Landmark Segmentation*: When triggered, the active landmark description is used as the text prompt for an open-vocabulary segmentation model:

$$S_t = \text{Seg}(O_t, l_i) = \{(\ell_{t,j}, S_{t,j})\}_{j=1}^{n_t}, \quad (7)$$

where each $S_{t,j}$ is an independent instance mask with label $\ell_{t,j}$. We instantiate this step with SAM 3 for text-conditioned concept segmentation [22].

4) *2D-to-3D Landmark Projection*: Each mask pixel with valid depth is back-projected using the camera intrinsics K and transformed into the map frame:

$$P_{\text{map},j}^l = T_{\text{map} \leftarrow \text{camera}} \Pi^{-1}(S_{t,j}, D_t, K), \quad j = 1, \dots, n_t. \quad (8)$$

After outlier rejection, projected observations are clustered. Sparse semantic memory stores landmark labels and cluster centers observed up to time t :

$$M_t^S = \{(\ell_j, \mu_j)\}_{j=1}^{N_t^S}. \quad (9)$$

where ℓ_j is the semantic label and μ_j the cluster center in the map frame. Each entry is a distinct landmark instance; same-label instances remain separate. Observations update the nearest compatible cluster within the matching threshold or create a new entry. Previously grounded landmarks remain in memory when no longer queried.

5) *Landmark-Conditioned Candidate Reasoning*: A landmark center is a semantic reference, not necessarily a traversable target. The VLM combines landmark positions with the active spatial relation to select among collision-checked frontier and local candidates (e.g., couches and a

counter define a left-turn passage). The path planner then executes the selected candidate on the geometric map.

F. Visual-Spatial Memory and Waypoint Execution

The geometric layer stores obstacles, free/unknown space, trajectory, and visited candidates; the sparse layer retains landmark labels and centers. Both are rendered with candidate IDs, while $E_t = \{O_{t-k+1}, \dots, O_t\}$ preserves recent appearance.

The decision input and selected waypoint are

$$X_t = \{M_t^G, M_t^S, E_t, s_{i_t}, q_t, C_t\}, \quad p_t^* = \text{VLM}(X_t) \in C_t. \quad (10)$$

The output identifies a valid candidate and provides a concise rationale. A classical planner computes a collision-free trajectory on the inflated map [23], separating semantic selection from metric execution.

IV. EXPERIMENTS

A. Benchmarks

We evaluate *SparseNav* on the continuous VLN benchmarks **R2R-CE** and **RxR-CE**. **R2R-CE** converts Room-to-Room trajectories into continuous navigation episodes in Habitat environments [1], [2], [24]. **RxR-CE** contains substantially longer trajectories and richer language descriptions, making it particularly useful for evaluating long-horizon instruction execution [25]. Experiments are conducted on the Val-Unseen splits to assess generalization to previously unseen environments. All experiments, including benchmark evaluation, ablations, and real-robot deployment, use GPT-5 as the VLM [26].

Following prior work, we use the standard VLN-CE metrics: Success Rate (SR), Success weighted by Path Length (SPL), Navigation Error (NE), Oracle Success Rate (OSR), and normalized Dynamic Time Warping (nDTW). We report SR, SPL, and NE on both benchmarks, OSR on **R2R-CE**, and nDTW on **RxR-CE**.

B. Results on Public Benchmarks

SparseNav achieves a Success Rate of 42.8% on **R2R-CE** Val-Unseen, outperforming several earlier zero-shot systems, including InstructNav, CA-Nav, AO-Planner, and DreamNav. On the more challenging **RxR-CE** benchmark, *SparseNav* achieves an SR of 40.7%. Table I also includes supervised methods for context. The reported SR of *SparseNav* is higher than those of NaVid and MapNav on both datasets, while ETPNav achieves a higher SR on both datasets and Dynam3D achieves a higher SR on **R2R-CE**. These are cross-paper comparisons rather than controlled reruns; differences in evaluated episodes, model backbones, and training data preclude a direct ranking.

Beyond SR, *SparseNav* achieves 35.2% SPL, 5.96 m NE, and 53.4% OSR on **R2R-CE**, as well as 24.1% SPL, 7.82 m NE, and 48.6% nDTW on **RxR-CE**.

The benchmark results demonstrate the effectiveness of *SparseNav*, but do not isolate the contribution made by sparse semantic perception.

C. Qualitative Analysis

Fig. 3 shows a navigation episode in which *SparseNav* follows the given instruction and stops at the specified destination. Representative grounding events are highlighted in the figure when instruction-relevant landmarks provide new spatial references for waypoint selection. Other stages proceed using previously grounded landmarks and geometric candidates without a new grounding event being depicted.

This episode illustrates how *SparseNav* coordinates persistent geometry with instruction-conditioned semantic perception: landmark memory supports reuse of acquired spatial evidence, while new grounding supplies additional evidence as needed. The agent thus follows the instruction without requiring semantic acquisition at every sub-instruction.

D. Ablation Studies

We evaluate semantic perception strategies, waypoint proposals, landmark grounding, instruction memory, semantic triggers, and map orientation through controlled ablations.

1) *Semantic Perception Strategy*: Table II separates semantic coverage from perception frequency. The dense-semantic continuous and on-demand settings reach 36.7% and 38.8% SR, respectively, whereas restricting queries to instruction-related landmarks gives 40.2% under continuous perception and 42.8% with on-demand perception. These controlled comparisons show that both the queried semantic scope and the trigger schedule affect navigation success; the *SparseNav* configuration combines instruction-related queries with on-demand grounding.

2) *Hybrid waypoint proposal*: Table III evaluates the contributions of frontier and local directional candidates. Local candidates alone achieve 29.1% SR, whereas frontier candidates alone achieve 35.2%. Adding a forward local candidate increases SR to 37.4%, and including left and right candidates further improves it to 42.8%. These results support combining frontier exploration with local motion options within explored free space.

3) *Landmark Grounding*: Table IV compares landmark localization strategies while keeping the remaining inputs fixed. Segmentation localization achieves an SR of 42.8%, compared with 40.2% for annotation-box localization, and reduces the reported localization error from 0.51 m to 0.12 m. The error is the Euclidean distance between the predicted 3-D cluster center and the true object center. These results suggest that more accurate landmark references can support waypoint selection.

4) *Instruction and Memory*: Table V evaluates instruction representation and memory on **RxR-CE**. With the full instruction, adding temporal memory increases the SR from 29.8% to 31.5%. Using sub-instructions further increases the SR to 33.2%, and adding progress tracking raises it to 35.0%. Adding persistent landmark memory increases SR from 35.0% to 40.7%, with instruction decomposition, temporal memory, and progress tracking unchanged. These results support the use of temporal context and explicit instruction progress for long-horizon navigation.

TABLE I

COMPARISON WITH SUPERVISED AND ZERO-SHOT VLN METHODS ON VAL-UNSEEN SPLITS. NE IS IN METERS; OTHER METRICS ARE PERCENTAGES.

Settings	Method	R2R-CE (Val-Unseen)				RxR-CE (Val-Unseen)			
		SR↑	SPL↑	NE↓	OSR↑	SR↑	SPL↑	NE↓	nDTW↑
Supervised	SASRA [27]	24.0	22.0	8.32	—	—	—	—	—
	Seq2Seq [1]	25.0	22.0	7.77	37.0	13.9	11.9	12.10	30.8
	CMA [1]	32.0	30.0	7.37	40.0	—	—	—	—
	NaVid [28]	37.4	35.9	5.47	49.1	23.8	21.2	8.41	—
	ETPNav [29]	57.0	49.0	4.71	65.0	54.8	44.9	5.64	61.9
	MapNav [11]	39.7	37.2	4.93	53.0	32.6	27.7	7.62	43.5
	Dynam3D [30]	52.9	45.7	5.34	62.1	—	—	—	—
Zero-shot	SmartWay* [31]	29.0	22.5	7.01	51.0	—	—	—	—
	OpenNav* [32]	19.0	16.1	6.70	23.0	—	—	—	—
	A2Nav* [33]	23.0	11.1	—	—	16.8	6.3	—	—
	InstructNav [15]	31.0	24.0	6.89	—	—	—	—	—
	AO-Planner [34]	25.5	16.6	6.95	38.3	22.4	15.1	10.75	33.1
	CA-Nav [16]	25.3	10.8	7.58	48.0	19.0	6.0	10.37	13.5
	DreamNav [20]	32.8	28.9	7.06	41.0	—	—	—	—
	SparseNav (Ours)	42.8	35.2	5.96	53.4	40.7	24.1	7.82	48.6

*Partly relies on simulator-labeled training data; zero-shot does not imply that every component is training-free. Protocols are not necessarily identical across rows. A dash denotes an unavailable value, not zero. Bold indicates the best performance among zero-shot methods.

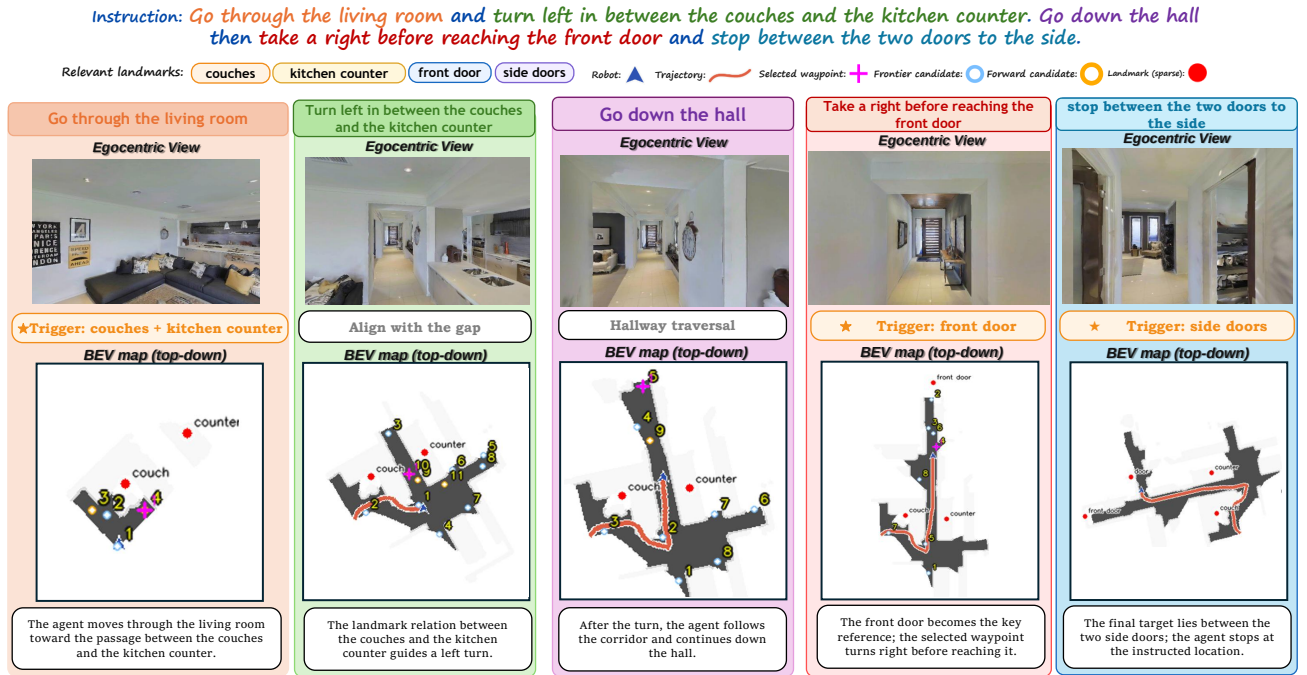


Fig. 3. Qualitative example of *SparseNav* following the given instruction. Each stage pairs an egocentric observation with its corresponding BEV map, illustrating selective landmark grounding, memory reuse, and waypoint selection.

TABLE II
SEMANTIC PERCEPTION STRATEGY.

Semantic strategy	SR↑
No semantic grounding	34.6
Dense semantics, continuous perception	36.7
Dense semantics, on-demand perception	38.8
Instruction-related semantics, continuous perception	40.2
SparseNav: instruction-related, on-demand perception	42.8

TABLE III
WAYPOINT CANDIDATE ABLATION

Frontier	Local-front	Local-left/right	SR↑
—	✓	✓	29.1
✓	—	—	35.2
✓	✓	—	37.4
✓	✓	✓	42.8

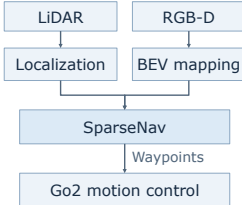
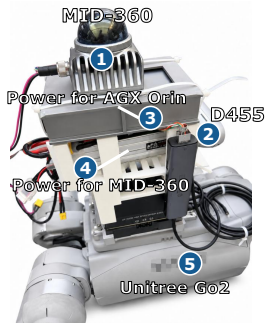
5) *Semantic Trigger Quality*: Table VI reports trigger quality on **R2R-CE**. A trigger is counted as correct when the queried object is visible and relevant to the current instruction; an appearing object followed by a segmentation call is therefore a successful trigger. Sub-instruction-aware triggering achieves the highest precision and recall, at 90.4% and 99.8%, respectively. For each active sub-instruction s_i , the semantic query is $Q_i = \text{Landmark}(s_i)$ and changes as

navigation progresses.

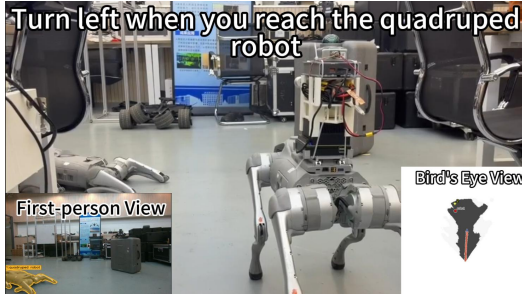
6) *BEV Coordinate Representation*: Table VII compares three BEV coordinate representations. The agent-centered but non-rotating configuration reaches 36.2% SR, while the agent-centered heading-up configuration reaches 42.8%, compared with 30.3% for global north-up. This comparison evaluates combined differences in map center and orientation rather than isolating orientation alone.

Instruction: **Go forward and turn left when you reach the quadruped robot, pass the humanoid robot then enter the next room and stop.**

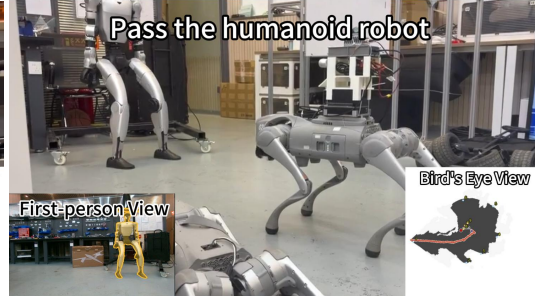
Platform architecture



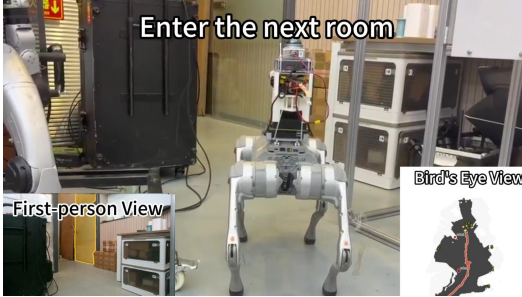
(a) Platform



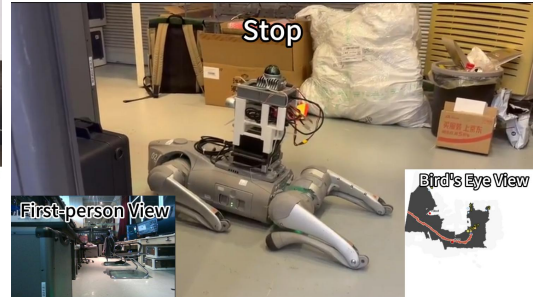
(b) Left turn



(c) Pass the humanoid robot



(d) Enter the next room



(e) Stop

Fig. 4. Real-robot platform and representative navigation sequence across the evaluated indoor environments: (a) hardware and navigation architecture; (b)–(e) representative execution snapshots showing the left turn, passing the humanoid robot, entering the adjacent room, and stopping.

TABLE IV

LANDMARK GROUNDING ABLATION.

Semantic localization	SR $\uparrow$	Landmark loc. error $\downarrow$
No localization	36.5	–
VLM-only localization	34.6	1.56 m
Annotation-box localization	40.2	0.51 m
Segmentation localization	42.8	0.12 m

TABLE V

INSTRUCTION AND MEMORY ABLATION ON RXR-CE.

Instruction representation	Temporal memory	Progress tracking	Persistent landmark memory	SR $\uparrow$
Full instruction	–	–	–	29.8
Full instruction	✓	–	–	31.5
Sub-instruction	✓	–	–	33.2
Sub-instruction	✓	✓	–	35.0
<i>SparseNav</i> full memory	✓	✓	✓	40.7

Taken together, these ablations support the central design of *SparseNav*: persistent geometry provides a spatial basis for navigation, while instruction-related semantic grounding combined with on-demand perception achieves the strongest SR among the evaluated semantic strategies. Hybrid waypoint proposals, precise landmark localization, and instruction memory further contribute to navigation performance.

E. Real-Robot Experiment

1) *Platform and environment*: Our platform is a Unitree Go2 quadruped equipped with an Intel RealSense D455 RGB-D camera for geometric mapping and landmark grounding, and a Livox MID-360 LiDAR for localization. We successfully deployed *SparseNav* on this platform without a prebuilt map and validated its effectiveness across multiple indoor environments.

2) *Navigation Task and Execution*: Across the evaluated scenarios, instructions define landmark-conditioned routes. A representative instruction is: “Go forward and turn left when

TABLE VI

SEMANTIC TOOL TRIGGER QUALITY ON R2R-CE.

Trigger	Precision $\uparrow$	Recall $\uparrow$
Periodic schedule	41.5	45.6
VLM visibility only	72.1	99.8
Sub-instruction-aware <i>SparseNav</i>	90.4	99.8

TABLE VII

BEV COORDINATE REPRESENTATION.

Map representation	SR $\uparrow$
Global north-up BEV	30.3
Agent-centered, north-up BEV	36.2
Agent-centered, heading-up BEV	42.8

you reach the quadruped robot, pass the humanoid robot then enter the next room and stop.” The quadruped robot referenced in the instruction is a scene landmark distinct from the Go2 executing the task. *SparseNav* tracks sub-instruction progress, selectively grounds relevant landmarks, and retains their positions in sparse semantic memory. The VLM selects waypoints from the hybrid geometric candidate set, and a path planner generates trajectories for execution.

3) *Real-World Deployment*: We successfully deployed *SparseNav* in real-world settings and validated its effectiveness across multiple indoor environments. Across these environments, instruction progress and grounded landmark positions guide waypoint selection, while the geometric map supports path planning. These experiments validate the effectiveness of integrating instruction-conditioned semantic grounding with geometric navigation on a real robot.

V. DISCUSSION AND CONCLUSION

We presented *SparseNav*, a training-free VLN framework that combines persistent geometric memory with instruction-conditioned sparse semantic grounding. Geometry represents traversable space, while the active instruction determines

which landmarks require grounding. Hybrid frontier and local waypoint candidates, accumulated landmark memory, and explicit instruction progress jointly support VLM-based navigation.

SparseNav achieves success rates of 42.8% on R2R-CE and 40.7% on RxR-CE, both on the Val-Unseen splits. Ablation studies show that instruction-related, on-demand semantic grounding achieves higher SR than the evaluated dense-semantic settings. Additional ablations support the contributions of hybrid waypoint proposals, precise landmark localization, instruction memory, and agent-centered map orientation. Successful deployment on a Unitree Go2 across multiple indoor environments further validates the effectiveness of the framework.

Together, these findings suggest that task-relevant sparse semantic grounding can be sufficient for semantic navigation: persistent spatial geometry can support effective instruction following without requiring a semantically complete scene map. The results highlight the potential value of selectively acquiring and reusing task-relevant landmarks. However, they do not establish that reduced visual clutter alone causes the performance gain, since selective grounding also changes the semantic evidence available to the planner.

VI. LIMITATIONS AND FUTURE WORK

Despite its effectiveness, *SparseNav* has several limitations. First, its rule-based waypoint generator restricts the VLM to a predefined set of frontier and local directional candidates. Although these candidates support both exploration and local maneuvers, they may omit waypoints that better match the instructed route, thereby limiting the flexibility of navigation. Second, selective grounding reduces segmentation overhead, but VLM inference remains a major computational bottleneck. In a sequential perception–reasoning–execution pipeline, waiting for waypoint decisions can delay robot motion and increase task completion time. Third, although *SparseNav* achieves competitive zero-shot results, its reported success rates on R2R-CE and RxR-CE remain below those of some navigation-trained methods. Differences in training data and evaluation settings prevent attributing this gap solely to the absence of navigation-specific training.

Future work will investigate learned waypoint proposals to provide more flexible, instruction-conditioned candidates while retaining geometric feasibility checks. Such extensions would be evaluated separately from the current training-free setting. We will also explore more efficient VLM inference and asynchronous planning and execution, allowing the robot to follow a validated local trajectory while preparing the next decision. This design will require mechanisms to detect stale plans and respond to changes in the observed environment. Finally, further real-world evaluations across more diverse layouts, landmark configurations, and instructions will assess the robustness and generalizability of the framework.

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